ISE 503: Data Management – Spring 2026

Course Project

Submission Deadline: Sunday, **May 3, 2026** by **11:59 PM (New York Time)** via **Brightspace**

Project Overview

This course project is focused on designing and implementing a **relational database system** that models real-world data and supports common transaction operations. You will use a relational database system such as **MySQL, PostgreSQL, SQLite, Oracle, or MariaDB** to create your database. The user interface and transaction handling can be implemented using **Java, JavaScript, JDBC**, or any equivalent technologies.

You are free to design your system based on your own structure and creativity, as long as the system is grounded in sound data modeling practices and uses SQL for database creation and interaction.

Team Requirements

- Maximum **team size: 3 students**
- Individual submissions are also allowed.
- Please submit your **team information** in the document with the link below:

  https://docs.google.com/spreadsheets/d/1TRINk5DTrAaqvLEZlPZfdZUAKZeJwQcOnuUmD8WjZvY/edit?gid=0#gid=0

Project Guidelines

The project should follow a typical database design workflow:

1. **Identify Entities**: Recognize key components (e.g., movies, candidates, trips).
2. **Define Attributes**: Capture relevant properties for each entity.
3. **Establish Relationships**: Define how entities interact and relate.
4. **Implement Constraints**: Define primary keys, foreign keys, domains, etc.
5. **Develop SQL Implementation**: Use SQL to build and manage the database.

6. **Build Application Interface (Optional)**: Design a basic interface to simulate transactions.

Project Topics (Choose One)

Project 1: Movie Database System

Design a system that manages data related to movies. This system should resemble platforms like **IMDb** or **Netflix**.

Suggested Entities:

- **Movie**: ID, Title, Release Date, Director, Producer, Cast, Duration, Genre, Rating
- **Actor**: ID, Name, Age, Awards
- **Director**: ID, Name, Age, Awards
- **Producer**: ID, Name
- **Distributor**: ID, Name, Address

Project 2: Job Search Portal

Create a system for an online job search portal, similar to **LinkedIn**, **Indeed**, or **CareerBuilder**.

Suggested Entities:

- **Job**: ID, Description, Salary, Employer ID
- **Employer**: ID, Name, CEO, Address
- **Candidate**: ID, Skill ID, Qualification, Experience
- **Skill**: ID, Name, Experience
- **Employee/Admin**: ID, Role, Join Date, Supervisor ID

Project 3: Travel Agency System

Design a system for a travel agency that supports booking, accommodation, and trip planning—similar to **Kayak** or **Expedia**.

Suggested Entities:

- **Location**: ID, City, State, Country
- **Transportation**: ID, Type (Flight, Car, Bus, Cruise), Class

- **Group**: ID, Size, Source, Destination, Mode, Purpose
- **Accommodation**: Type, Rate, Facilities, Discount
- **Admin/Employee**: ID, Role, Join Date, Supervisor ID
- **Car Rental**: Confirmation ID, Car Type, Rent
- **Flight**: Number, Carrier, Source, Destination, Class, Fare
- **Cruise**: Number, Source, Destination, Fare
- **Passenger**: ID, Name, Gender, Age
- **Payment**: Type, Card Number, Expiry Date
- **Reviews**: Rating, Text, Passenger ID, Group ID

You may extend these schemas with additional relevant entities like activities (e.g., scuba diving, hiking) for enhanced functionality.

Deliverables

- **Relational Schema** with proper constraints
- **SQL Scripts** to:
  - Create and populate the database
  - Answer **10 complex queries** involving joins
- **Sample Data**: At least 30 records per table (on average)
- (Optional) Basic user interface or demo

Academic Integrity

All work must be original and created by your team. Any evidence of plagiarism or code sharing will be treated seriously and reported under university policy.

If you have any questions or would like to propose a custom project idea, feel free to reach out.

Good luck, and enjoy building your systems!